

Improving Conversational Recommendation Systems with Natural Language Understanding and Large Language Models

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Abstract

Purpose - This study aims to investigate and improve on existing film recommendation systems by giving users an explanation for how the recommendation was made and to permit the user to provide feedback on recommendations in a conversational manner, employing natural language understanding (NLU) and large language models (LLMs) to alter and tune the recommendation system to emphasise weightings on certain attributes.

Originality/Value - This project takes a novel approach on the use of conversational AI to attempt to improve on current recommendation systems, particularly in the domain of film recommendations. By leveraging natural language understanding (NLU) and large language models (LLMs), the system aims to provide more accurate and personalized recommendations. Additionally, the system will offer explanations for its recommendations and allow users to provide feedback in a conversational manner, which will be used to fine-tune the recommendation algorithm.

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1 Introduction

1.1 Document Structure

The following document will discuss:

- The background of the project
- The purpose of the project
- The scope of the project
- The aims and objectives of the project
- The requirements of the project
- The technologies, tools and datasets used in the project
- The evaluation methodology used in the project

1.2 Project Diary

The project diary is available [here](#).

1.3 Project Purpose

The purpose of this project is to investigate the feasibility of using Natural Language Processing (NLP) technologies in addition to Large Language Models to improve on existing conversational recommendation systems.

This will be through the use of 'chatbots' which are able to understand and respond to natural language queries. The usage of chatbots is to provide a more natural and intuitive way for users to interact with the recommendation system. It is hoped that this will lead to more accurate and relevant recommendations being made to users, while also making the user feel heard and understood.

The system will provide recommendations to the user, provide explanations for the recommendations made, and allow the user to provide feedback on the recommendations in a conversational manner. This feedback will be used to fine-tune the recommendation algorithm, with the aim of providing more accurate and personalized recommendations to the user. The chatbots will also be used to guide the user to explore and understand their preferences better, and to reduce the cold start problem in recommendation systems.

1.4 Project Scope

The project will focus on the following areas:

- Implementing a chatbot that can interact with users and build a user profile (e.g. user document matrix).
- Implementing large language models to generate natural, free flowing conversations with the user.
- Explaining the recommendations made by the system to the user.
- Investigate the opportunities in guiding the user to explore and understand their preferences better.
- Gathering both explicit and implicit feedback from the user to improve recommendations.
- Working with the user to refine their preferences in a 'session' to provide a set of film recommendations, for that session (e.g. films for a movie night in).
- Evaluating the performance of the chatbot in terms of reducing the cold start problem
- Evaluating the performance of the chatbot in terms of recommendation quality and user satisfaction

1.5 Background and project topic discussion

Currently, recommendation systems typically use either or both collaborative filtering content-based filtering to make recommendations to users. These systems are limited in their ability to understand the user's preferences and provide relevant recommendations. Often, users are required to provide explicit feedback on items they like or dislike in order to receive accurate recommendations, and even then, the recommendations may not be relevant or useful to the user.

The use of chatbots in recommendation systems is a relatively new area of research, but it has the potential to revolutionize the way recommendations are made to users. By using chatbots, recommendation systems can provide a more natural and intuitive way for users to interact with the system, gathering explicit feedback from the user in a more natural way for adjusting recommendations. The aim is to provide a method that is non-intrusive in its feedback gathering, where the user feels they are having a conversation with the system, rather than being asked to fill out a survey.

Recent advancements in conversational AI have shown significant potential in enhancing recommendation systems. According to a survey[13] on the current state of conversational recommender systems, integrating natural language processing and dialogue management techniques can improve user satisfaction and engagement. These systems can understand user preferences through natural conversations, making the recommendation process more dynamic and personalized.

I found a project in this area an interesting proposition, as it seeks to address a common problem that I, and many others face when trying to find something to watch on Netflix or Amazon Prime. While I expect this project may not be viable for a commercial use due to its cost, I believe a

proof of concept is viable and could show the potential of this technology. I seek to investigate and explore the possibilities of using large language models in conversational recommendation systems, and to evaluate the system compared to previous recommendation systems.

1.6 Project Definition

The project will aim to address the following research questions:

- Can chatbots be used to improve the quality of recommendations made to users?
- Can this system explain the recommendations made to the user?
- Can natural, free flowing conversations that LLMs can have with users be integrated into information retrieval systems?
- Can chatbots be used to guide users to explore and understand their preferences better?
- Can chatbots be used to reduce the cold start problem in recommendation systems?
- Can chatbots be used to improve user satisfaction through refining user preferences with this recommendation systems?

I chose a project that is a proof of concept, which has both development and research aspects. This project is a novel area of research, and I believe it has the potential to show the benefits of using chatbots in recommendation systems. It involves both research and development, as I will need to investigate the current state of the art in recommendation systems and NLP, and then implement a system that integrates these technologies.

2 Goals and Requirements

The goal of this project is to investigate the feasibility of using Natural Language Processing (NLP) technologies in addition to Large Language Models to improve on existing conversational recommendation systems. This will be through the use of 'chatbots' which are able to understand and respond to natural language queries.

The usage of chatbots is to provide a more natural and intuitive way for users to interact with the recommendation system. It is hoped that this will lead to more accurate and relevant recommendations being made to users, while also making the user feel heard and understood.

Requirements:

- It will attempt to leverage state-of-the-art NLP techniques to process and understand user inputs.
- It will also explore the integration of Large Language Models to generate coherent and contextually appropriate responses.
- The project will aim to develop a user-friendly chatbot interface that can handle a variety of conversational scenarios within the conversational recommendation domain.
- Additionally, it will seek to evaluate the effectiveness of these technologies in enhancing the accuracy and relevance of recommendations.

3 Preliminary Literature Review

Before commencing the project, I looked at research that has been done in this field previously, to understand the current state of the art and to identify gaps in the research that had an opportunity for me to contribute. Conversational recommender systems have only relatively recently had a rise in popularity.

3.1 Existing Approaches

Collaborative filtering is a popular approach to recommendation systems, which is based on the idea that users who have similar preferences in the past will have similar preferences in the future. This approach is widely used in recommendation systems such as Netflix, Amazon, and Spotify, where users are recommended items based on the preferences of other users who have similar tastes.

Content-based filtering is another approach to recommendation systems, which is based on the content of the items. This approach is used in systems such as YouTube and Google News, where users are recommended items based on the content of the items they have interacted with in the past.

Hybrid recommender systems combine collaborative and content-based filtering to provide more accurate and personalized recommendations. These systems are used in a variety of domains, including e-commerce, entertainment, and social media, where users are recommended items based on their past behaviour and the content of the items.

3.2 *Papers*

3.2.1 **Conversational Recommender Systems**

Conversational recommendation systems (CRSs) represent a rapidly evolving intersection of artificial intelligence, natural language processing, and user interaction design, aimed at enhancing personalized experiences across diverse domains such as e-commerce, entertainment, and customer service. By facilitating engaging dialogues between users and machines, CRSs provide dynamic and context-aware recommendations tailored to individual preferences, effectively addressing the complexity of choices available in today’s digital landscape.

3.2.2 **Leveraging Large Language Models in Conversational Recommender Systems**

Recent developments[1] in large language models (LLMs) have significantly improved the capabilities of conversational recommenders. Notably, studies have demonstrated that LLMs can serve as effective zero-shot conversational recommenders, allowing systems to generate recommendations based on limited prior knowledge of user preferences. These advancements are complemented by the integration of tools and external knowledge bases, enabling systems to perform complex tasks and provide more accurate recommendations

3.2.3 **Data scarcity and training cost**

Data scarcity remains a significant challenge in training effective Conversational Recommender Systems (CRSs). To address this issue, techniques such as model distillation and fine-tuning can be employed to enhance model performance while reducing computational costs.

In the paper by Hsieh et al. [11], the authors propose a step-by-step distillation approach. This method involves training a smaller model to mimic the behavior of a larger model, followed by fine-tuning the smaller model on the target task. This approach not only outperforms the larger model on the target task but also proves to be more computationally efficient.

Another strategy to mitigate data scarcity is the generation of synthetic data using a larger pre-trained model. This technique leverages the capabilities of a large language model to create additional high quality training examples, which can then be used to train a smaller, more efficient model. By augmenting the training dataset with synthetic data, the performance of the conversational recommender system can be improved without the need for extensive real-world data collection. This method is particularly useful in scenarios where obtaining large amounts of labeled data is challenging or costly. For instance, in this project, it is not feasible for me to have a large number of users interacting with the system to generate training data.

3.2.4 **Evaluation**

The evaluation of conversational recommender systems (CRSs) is a multifaceted challenge that encompasses various dimensions such as recommendation accuracy, user satisfaction, and interaction quality. Traditional metrics like precision, recall, and F1-score are often employed to measure the accuracy of recommendations. However, these metrics alone are insufficient to capture the interactive and dynamic nature of CRSs.

Recent research has emphasized the importance of user-centric evaluation metrics, including user satisfaction, engagement, and perceived usefulness. User studies and A/B testing are commonly used to gather qualitative feedback and assess the overall user experience. Additionally, metrics such as dialogue coherence, response relevance, and interaction length are considered to evaluate the quality of the conversational interactions.

Moreover, the use of simulation environments and user simulators has gained traction[12] as a means to conduct large-scale evaluations without the need for extensive real-world user interactions.

These simulators can model user behavior and preferences, providing a controlled setting to test and refine CRSs.

Despite these advancements, papers outline that there remains a need for standardized evaluation frameworks that can comprehensively assess the performance of CRSs across different domains and use cases.

3.2.5 Explainability

In conversational recommendation systems, explainability is crucial for building trust with users. The system must be able to clearly explain why a particular recommendation was made and how it arrived at that conclusion.

Current approaches to explainability in recommendation systems, such as those discussed by Guo et al. [7], or Xu et al. [15] focus on providing transparent and understandable justifications for recommendations. These approaches often involve breaking down the recommendation process into comprehensible steps that can be communicated to the user. For instance, a system might explain that a movie recommendation is based on the user's previous preferences for a specific genre or the popularity of the movie among similar users.

Recent advancements in natural language processing have enabled more sophisticated explainability mechanisms. NLP techniques can be used to generate natural language explanations that are easy for users to understand. Additionally, dialogue management systems can handle follow-up questions from users, providing further clarification and enhancing the overall transparency of the recommendation process.

In this project, I aim to incorporate these explainability techniques to ensure that users feel confident and informed about the recommendations they receive. By leveraging both content-based and collaborative filtering explanations, as well as utilizing NLP for natural language justifications, we can create a conversational recommendation system that not only provides accurate recommendations but also builds trust and engagement with users.

3.2.6 Critiquing-based recommendation systems

My project touches in on critiquing-based recommendation systems, where the user is asked to provide feedback on the recommendations they are given, and the system uses this feedback to refine the recommendations. Current recommendation systems are limited in that they are only based on static attribute values (for example, in a digital marketplace, attributes such as a digital camera's screen size, effectiveness pixels, optical zoom).

One paper, Chen et al[4], proposes a sentiment-integrated critiquing approach, for helping users to formulate and refine their preferences, improving the user's product knowledge, preference certainty, decision confidence, perceived information usefulness, and purchase intention.

This can be integrated into a conversational recommender system, where the user sentiment can be used to refine the recommendations given to the user (e.g. the system might be picking up that a user isn't interested in a particular genre of movie, but isn't explicitly saying so, so the system can ask the user for feedback on the genre of movie they are being recommended).

3.2.7 Context-aware recommendation

While not directly applicable to my project, this paper from Carchiolo et al[3] proposes a context-aware recommendation network, which can be used to find experts in a particular field. If I have time, I plan to experiment with identifying 'expert' users and using their preferences to improve the recommendations given to other users. For example a user with a lot of experience in a particular genre of movie might be able to provide more accurate feedback on the recommendations they are given, which can then be used to improve the recommendations given to other users.

This is basically building on normal information retrieval systems, which already take into account problems with low quality users - e.g. ones with one or two reviews in total.

4 Planned System Outline

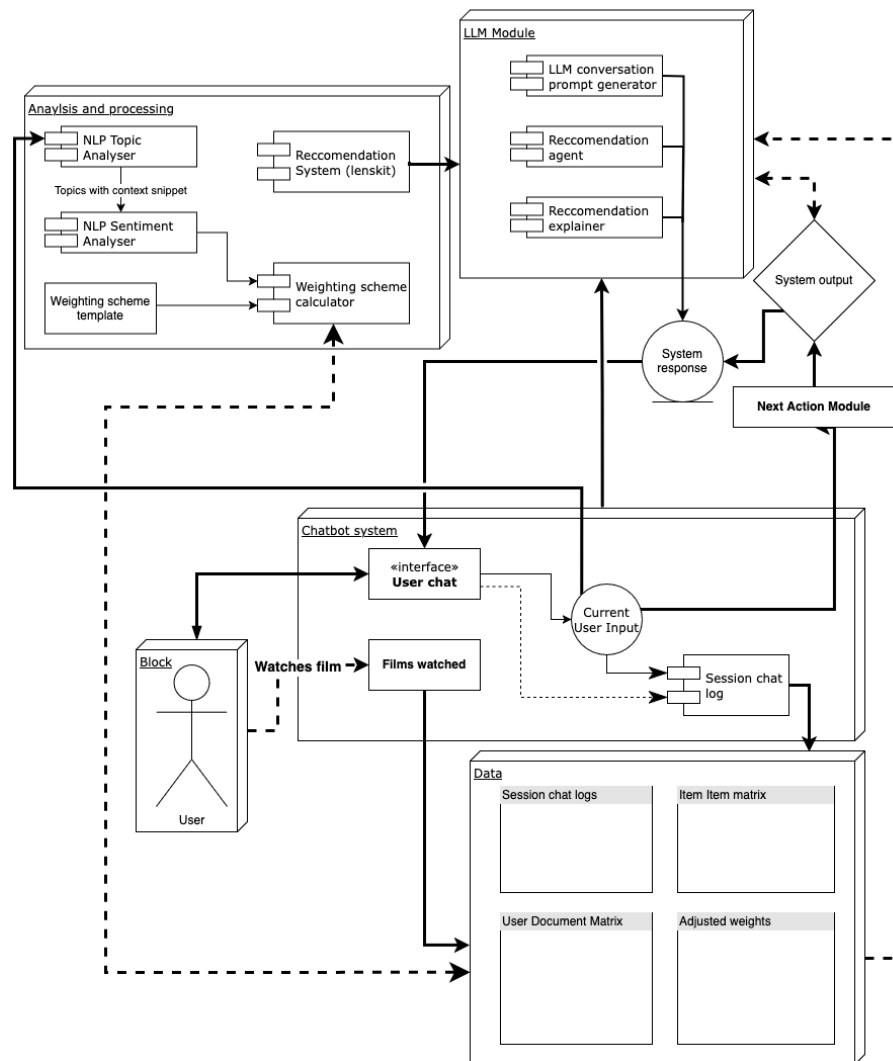


Figure 1: System High-Level Architecture

The system will be built with a multi-agent approach, with each agent responsible for a different aspect of the system.

I will be building the project in steps, where I am keeping each component as modular as possible to allow for easy integration and testing.

The system will be built with the following components:

- **Chat interface:** The user interface for the user to interact and converse with the system.
- **NLP Module:**
 - Topic Extraction:** Extracting topics and features from the user's messages.
 - Sentiment Analysis:** Extracting topic sentiment from the user's messages.
- **Recommendation Module:**
 - RecSys:** Collaborative and content-based filtering recommendations.
 - Decision Values:** Provides result values for a given recommendation.
- **LLM Module:** Using Large Language Models to converse with the user.
 - User Preference Exploration:** Help guide the user to explore their preferences.
 - Rec Explainability:** Explaining how and why a recommendation was made.
- **Sentiment Weighting Calculator:** Takes sentiment analysis and adjusts users weighting scheme accordingly.

- **Next Action Module:** Based on the user's messages and the system's current state, the module decides on what action to do next.

5 Technologies

This section provides an overview of the various software, tools, and datasets used in the project.

5.1 Software

- **Python:** The majority of this project will be written in Python. I chose python because it is a versatile language with a wide range of libraries and frameworks for machine learning and data science.
- **TensorFlow:** An open-source machine learning framework used for building and training neural networks. I will be using TensorFlow to build and train the large language models.
- **Jupyter Notebook:** An interactive computing environment that enables users to create and share documents containing live code, equations, visualizations, and narrative text.
- **Git & Github:** I will be using Git for version control and Github for hosting the project repository.

5.2 Libraries & Tools

- **spaCy:** An open-source software library for advanced natural language processing in Python. I will be using spaCy for tokenization, lemmatization, and named entity recognition. [10]
- **Twitter-roBERTa-base for Sentiment Analysis:** NLP RoBERTa model for sentiment analysis. I will be using this model to extract sentiment from user responses. [2]
- **Hugging Face Transformers:** I will be using the hugging face transformers library to load and fine-tune large language models. The library has a wide range of pre-trained models that can be fine-tuned on custom datasets. [14]
- **LensKit:** A toolkit for building, researching, and studying recommender systems. I will be using LensKit for building and evaluating the conversational recommender system. [6]

5.3 Datasets

- **MovieLens 32M:** This dataset[8] is a stable benchmark dataset. It has 32 million ratings and two million tag applications applied to 87,585 movies by 200,948 users. Collected on 10/2023, released 05/2024. I will be using this dataset for training and evaluating the conversational recommender system in terms of the quality of recommendations.
- **INSPIRED:** This is a dataset[9] of 1,001 human-human dialogs for movie recommendation with measures for successful recommendations. The dataset includes annotations based on social science theories to understand how humans make recommendations in communication. Analysis shows that sociable recommendation strategies, such as sharing personal opinions or communicating with encouragement, more frequently lead to successful recommendations. Models trained with these strategy labels outperform baseline models in both automatic and human evaluations. This work aims to build sociable recommendation dialog systems grounded in social science theories. I will be using this dataset to train and evaluate the conversational recommender system in terms of both the quality of recommendations and the quality of the conversation.
- **The Movie Database (TMDB) API:** TMDB[5] is a community built movie and TV database. The database contains information on movies, TV shows, actors, and various other pieces of data. I'm using it as it contains metadata on movies, such as genres, actors, directors, and more. This data will be used to provide information on movies to the user, and to build into recommendations.

6 Evaluation Methodology

Due to the complex nature of conversational recommender systems, I will be evaluating the system in two different approaches: **system-centric** and **user-centric** evaluations.

The system-centric evaluation will focus on the performance of the system in terms of recommendation quality, while the user-centric evaluation will focus on the user experience and satisfaction with the system.

The evaluation of CRSs has many challenges, including the need for user involvement, fairness in recommendations, and the necessity for multi-metric assessments that capture various performance dimensions.

6.1 *System-Centric Evaluation*

The system-centric evaluation will focus on evaluating the performance of the conversational recommender system. It will focus on the algorithmic accuracy and metrics such as precision and recall, typically relying on historical data.

6.1.1 Evaluation Metrics

I will be judging the system performance based on the following metrics:

- **Recommendation Quality:** How good are the recommendations made by the system?
- **Recommendation Accuracy:** How accurate are the systems recommendations?
- **Recommendation Diversity:** How diverse are the systems recommendations?
- **Recommendation Novelty & Serendipity:** How novel and surprising are the recommendations made by the system?
- **Recommendation Coverage:** How many correct items can the system recommend?
- **Recommendation Explainability:** How well can the system explain its recommendations?

I will be using the evaluation metrics from the LensKit library [6] to evaluate the performance of the conversational recommender system. This will be both prediction accuracy metrics and top-N recommendation metrics.

6.2 *User-Centric Evaluation*

The user-centric evaluations prioritize user engagement and satisfaction, which will often require direct feedback from users.

6.2.1 User Feedback Illicitation

User-centric evaluations require direct feedback from the users interacting with the system. This feedback can be explicit, with users providing ratings or preferences or implicit, inferred from user behaviour such as clicks or dwell time.

For this project, I don't believe the scope of my project will allow for a full user study, but I will aim to gather feedback from users in a explicit manner, such as through live user testing and post interaction surveys.

The scope will also limit me in implementing implicit feedback mechanisms, such as tracking user interactions with the system. Some implicit feedback mechanisms I could implement include tracking how many movies a user watches after a recommendation.

6.2.2 Evaluation Metrics

I will be judging the user experience based on the following metrics:

- **User Satisfaction:** How satisfied are users with the system?
- **User Engagement:** How engaged are users with the system?
- **User Retention:** How many users would return to use the system again?
- **User Feedback:** What feedback do users provide about the system?

- **User Usage Manner:** How did the user interact with the system?
- **User Preferences:** How well does the system understand user preferences?
- **User Recommendations:** How well does the system recommend items to users? Does the user think the recommendations are relevant?
- **User Conversations:** How well does the system engage in conversations with users? Does the user feel listened to and understood?

7 Planned Steps

The project will be broken down into the following stages:

- **Stage 1: Literature Review** - A critical review of literature related to conversational recommender systems and large language models.
- **Stage 2: Prototype Development** - Developing prototype systems for each of the conversational recommender system components.
- **Stage 3: Data Preprocessing** - Preprocessing the datasets to prepare them for training and evaluation.
- **Stage 4: System Integration** - Integrating proof of concept level implementations of the conversational recommender system components into a single system.
- **Stage 5: Model training and fine tuning** - Training the LLM on the data and fine tuning it for the conversational recommender system.
- **Stage 6: Distillation** - Distillation of the model to reduce the size of the model, while maintaining its performance.
- **Stage 7: Model Evaluation** - Evaluating the performance of the conversational recommender system using various metrics.
- **Stage 8: Model Optimization** - Optimizing the conversational recommender system to improve its performance.
- **Stage 9: User Testing** - Testing the conversational recommender system with real users to gather feedback.
- **Stage 10: Final Evaluation** - Evaluating the final version of the conversational recommender system and comparing it to existing systems.
- **Stage 11: Documentation** - Writing the final project report and preparing for the project presentation.

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